

● HARD

● PROBLEM-SOLVING AND DATA ANALYSIS

● ANSWER KEY

Ratios, rates, proportional relationships, and units

02

7 answers · Rationale-rich · Teach from doc

Q1**.0625**

Accept also: 1/16

The correct answer is .0625 . It's given that $\frac{4a}{b} = 6.7$ and $\frac{a}{bn} = 26.8$. The equation $\frac{4a}{b} = 6.7$ can be rewritten as $4\frac{a}{b} = 6.7$. Dividing both sides of this equation by 4 yields $\frac{a}{b} = 1.675$. The equation $\frac{a}{bn} = 26.8$ can be rewritten as $\frac{a}{b} \frac{1}{n} = 26.8$. Substituting 1.675 for $\frac{a}{b}$ in this equation yields $1.675 \frac{1}{n} = 26.8$, or $\frac{1.675}{n} = 26.8$. Multiplying both sides of this equation by n yields $1.675 = 26.8n$. Dividing both sides of this equation by 26.8 yields $n = 0.0625$. Therefore, the value of n is 0.0625 . Note that .0625, 0.062, 0.063, and 1/16 are examples of ways to enter a correct answer.

Q2**C****C.** $\frac{3x}{17}$

Choice C is correct. It's given that in a scale model, a length of 1 inch is equivalent to a length of $\frac{17}{3}$ feet in the actual office building. This means that the ratio of a length, in feet, in the actual office building to the corresponding length, in inches, in the model is $\frac{17}{3}$ to 1. Therefore, the length, in inches, in the model corresponding to a length in the actual office building of x feet can be found by dividing x by $\frac{17}{3}$. Dividing by $\frac{17}{3}$ is equivalent to multiplying by $\frac{3}{17}$, so the corresponding length, in inches, in the model in terms of x can be represented by the expression $\frac{3x}{17}$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect. This expression represents the corresponding length, in inches, in the model in terms of x for a model where 1 inch in the model is equivalent to a length of $\frac{3}{17}$, not $\frac{17}{3}$, feet in the actual building.

Q3**4176**

The correct answer is 4,176. It's given that the side length of the larger square is 3 times the side length of the smaller square. This means that the area of the larger square is 3^2 , or 9, times the area of the smaller square. If the area of the smaller square is represented by x , then the area of the larger square can be represented by $9x$. Therefore, the flat surface of the two adjacent squares has a total area of $x + 9x$, or $10x$. It's given that an electric field with strength 29.00 volts per meter passes uniformly through this surface and the total electric flux of the electric field through this surface is 4,640 volts · meters. Since it's given that the electric flux is the product of the electric field's strength and the area of the surface, the equation $29.0010x = 4,640$, or $290x = 4,640$, can be used to represent this situation. Dividing each side of this equation by 290 yields $x = 16$. Substituting 16 for x in the expression for the area of the larger square, $9x$, yields 916, or 144, square meters. Since the area of the larger square is 144 square meters, the electric flux, in volts · meters, of the electric field through the larger square can be determined by multiplying the area of the larger square by the strength of the electric field. Thus, the electric flux is 144 square meters $\frac{29.00 \text{ volts}}{\text{meter}}$, or 4,176 volts · meters.

Q4**157.8**

Accept also: 789/5

The correct answer is 157.8. It's given that an area of 46.00 square nautical miles is equivalent to k square kilometers. Since 1 nautical mile is equal to 1.852 kilometers, it follows that an area of 1 square nautical mile is equivalent to $(1.852)^2$, or 3.429904, square kilometers. Multiplying 46.00 by 3.429904 yields 157.775584 square kilometers. Rounding this value to the nearest tenth yields 157.8. Therefore, the value of k is 157.8.

Q5**D****D.** 13,505,536

Choice D is correct. Since the number of yards in 1 mile is 1,760, the number of square yards in 1 square mile is $(1,760)(1,760) = 3,097,600$. Therefore, if the area of the town is 4.36 square miles, it is $4.36(3,097,600) = 13,505,536$, in square yards.

Choice A is incorrect and may result from dividing the number of yards in a mile by the square mileage of the town.

Choice B is incorrect and may result from multiplying the number of yards in a mile by the square mileage of the town.

Choice C is incorrect and may result from dividing the number of square yards in a square mile by the square mileage of the town.

Q6**B****B.** 3.83

Choice B is correct. Since 1 mile is equal to 1,760 yards, 1 square mile is equal to $1,760^2$, or 3,097,600, square yards. It's given that the park has an area of 11,863,808 square yards.

Therefore, the park has an area of 11,863,808 square yards $\frac{1 \text{ square mile}}{3,097,600 \text{ square yards}}$, or $\frac{11,863,808}{3,097,600}$ square miles. Thus, the area, in square miles, of the park is 3.83.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the square root of the area of the park in square yards, not the area of the park in square miles.

Choice D is incorrect and may result from converting 11,863,808 yards to miles, rather than converting 11,863,808 square yards to square miles.

Q7**B****B. 16.3**

Choice B is correct. It's given that the speed of a vehicle is increasing at a rate of 7.3 meters per second squared. It's given to use 1 mile = 1,609 meters. There are 60 seconds in 1 minute; therefore, 60^2 or 3,600 seconds squared is equal to 1 minute squared. It follows that the rate of 7.3 meters per second squared is equivalent to $\frac{7.3 \text{ meters}}{1 \text{ second squared}} \cdot \frac{1 \text{ mile}}{1,609 \text{ meters}} \cdot \frac{3,600 \text{ seconds squared}}{1 \text{ minute squared}}$, or approximately 16.33 miles per minute squared. The rate, in miles per minute squared, rounded to the nearest tenth is 16.3.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

Generated 2026-05-29 · cb-educator-question-bank